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Electrochemical study of the effect of Al^{3+} on the stability and performance of Cu-based ATRP catalysts in organic media

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ABSTRACT

Electrochemically-mediated atom transfer radical polymerization (*e*ATRP) has attracted a great deal of attention as one of the most powerful methods of synthesis of polymers with well-defined architecture and low dispersity. The process is more cost-effective and simpler if performed in undivided cells with cheap sacrificial anodes such as aluminum. In this connection, knowledge of the effect of Al^{3+} ions released from the anode on the stability and performance of typical Cu catalysts is highly desired. In this study, the stability of Cu complexes with tris(2-(dimethylamino)ethyl)amine (Me_6TREN) and tris(2-pyridylmethyl)amine (TPMA), was investigated in DMF, DMSO, and MeCN by cyclic voltammetry, UV-vis-NIR spectroscopy and controlled-potential *e*ATRP. Both ligands form complexes with Al^{3+}, but in DMF and DMSO, TPMA binds more strongly to Cu^{2+} and Cu^{+} than Al^{3+}, whereas the opposite is true for Me_6TREN in all tested solvents. Therefore, *e*ATRP of *n*-butyl acrylate catalyzed by $[Cu^{I}TPMA]^{+}$ in DMF or DMSO was not affected by Al^{3+} ions. In contrast, when Me_6TREN was used, the reactions could not proceed to high conversions because Al^{3+} ions displace Cu^{2+} from the ligand. To suppress the effect of Al^{3+}, use of excess ligand and/or Br^{-} with respect to Cu^{II} proved to be efficient.

1. Introduction

Over the past few decades, a great deal of effort was made to develop reversible deactivation radical polymerization (RDRP) methods for the synthesis of polymers with a well-defined structure, low dispersity and predetermined composition [1]. Various RDRP techniques such as nitroxide mediated polymerization (NMP) [2] organometallic mediated radical polymerization (OMRP) [3,4], reversible addition-fragmentation chain-transfer (RAFT) [5,6], and atom transfer radical polymerization (ATRP) [7,8] have been developed but the last has attracted most attention because of its robustness, simplicity and applicability to a wide range of monomers and reaction media [9].

The mechanism of traditional copper-catalyzed ATRP is shown in Scheme 1 (region delimited by the dashed line) [10,11]. $[Cu^{I}L]^{+}$ (L = polydentate amine ligand) activates the dormant species (an activated alkyl halide, R-X, used as an initiator or a halogen-capped polymeric chain, P_n–X) by a reductive cleavage of the C-X bond, resulting in the formation of propagating radicals and the Cu complex at a higher oxidation state, i.e. $[XCu^{II}L]^{+}$, which acts as a deactivator. In the deactivation step, $P_n^{\bullet}$ is converted back to the dormant species P_n-X while $[XCu^{II}L]^{+}$ is reduced to $[Cu^{I}L]^{+}$. The equilibrium is well-shifted to the left ($K_{ATRP} << 1$) so that the propagating radicals are always present in solution at a very low concentration, thus disfavoring radical-radical termination reactions [12]. Polymer chains grow as they are intermittently activated and deactivated, but most of the time they stay in the dormant state. The rate of polymerization, R_p, depends on the rate of radical propagation (k_p) as well as on the concentrations of all involved species and the ATRP equilibrium constant, as shown in Eq. (1).

$$R_P = k_p[M]_0[P_n^{\bullet}] = k_p[M]_0 K_{ATRP}\frac{[RX]_0 \cdot C_{[Cu^{I}L]^{+}}}{C_{[XCu^{II}L]^{+}}} \quad (1)$$

Traditional ATRP requires starting with air sensitive Cu^{I} at relatively high concentrations with the obvious trouble of successive workup for accurate polymer purification. Various advanced methods of ATRP aiming to avoid starting with Cu^{I} and reducing the catalyst load to ppm levels have been developed and are nowadays commonly used [13]. These methods are based on the use of air-stable Cu^{II} and (re)generation of Cu^{I} in situ by different reduction processes. These methods include initiators for continuous activator regeneration (ICAR) ATRP [14,15], activators regenerated by electron transfer (ARGET) ATRP [15,16], supplemental activator and reducing agent (SARA) ATRP [17–19], electrochemically mediated ATRP (*e*ATRP) [20–23], photochemically mediated ATRP (photoATRP) [24–26] and mechanoATRP [27–29].

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